

Set A

Q1.

$$\text{Total Pixels per frame} = 2560 \times 1440 \\ = \boxed{3686400}$$

$$\text{Time per frame} = \frac{1}{75} = 13.33 \text{ ms}$$

In 1ms GPU can render = 130,000 pixels

$$\therefore \text{In } 13.33 \text{ ms " " " } = 130,000 \times 13.33 \text{ pixels} \\ = \boxed{1732900} \text{ pixels}$$

$$\boxed{1732900 < 3686400}$$

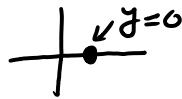
So, GPU cannot.

$$\text{Frame Size} = 3686400 \times 30 \times \frac{1}{8 \times 10^6} \text{ MB}$$

$\uparrow$
bit to byte

Q2.

a) $3x + 4y - 30 = 0$
 $\Rightarrow 3x + 4 \times 0 - 30 = 0$
 $\Rightarrow x = 10$



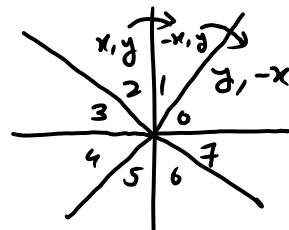
Starting: (10, 0) $dx = -5$ (-) $|dy| > |dx|$

Ending: (5, 7) $dy = 7$ (+) Zone: 2

Zone 2 $\rightarrow$ Zone 0

(10, 0) $\rightarrow$ (0, -10)

(5, 7) $\rightarrow$ (7, -5)



$$dx = 7$$

$$dy = 5$$

$$d_{init} = 2dy - dx = 3$$

$$\Delta d_{NE} = 2(dy - dx) = -4$$

$$\Delta d_E = 2dy = 10$$

$$d_{init} = 3 ; \Delta d_{NE} = -4 ; \Delta d_E = 10$$

x	y	d	NE/E	d_{update}	Zone 2 Pixels
0	-10	$3 > 0$	NE	$3 + \Delta d_{NE} = -1$	(10, 0)
1	-9	$-1 < 0$	E	$-1 + \Delta d_E = 9$	(9, 1)
2	-9	$9 > 0$	NE	$9 + \Delta d_{NE} = 5$	(9, 2)
3	-8	$5 > 0$	NE	$5 + \Delta d_{NE} = 1$	(8, 3)
4	-7	$1 > 0$	NE	$1 + \Delta d_{NE} = -3$	(7, 4)
5	-6	$-3 < 0$	E	$-3 + \Delta d_E = 7$	(6, 5)

$x = -Y, y = X$

b) It is because, multiplying the decision parameter by 2 removes the fractional value, but it does not change the sign. The decision (NE/E) remains the same.

Set B

Q1.

$$\text{Total Pixels per frame} = 1600 \times 900 \\ = 1440,000$$

$$\text{Time per frame} = \frac{1}{75} = 13.33 \text{ ms}$$

In 1ms GPU can render = 2500,000 pixels

$$\therefore \text{In } 13.33 \text{ ms " " " } = 2500,000 \times 13.33 \text{ pixels} \\ = 33325,000 \text{ pixels}$$

$$33325,000 > 1440,000$$

So, GPU can.

$$\text{Frame Size} = 1440,000 \times 24 \times \frac{1}{8 \times 10^6} \text{ MB}$$

↑
bit to byte

Q2.

$$\begin{aligned} a) \quad 4x + 3y - 30 &= 0 \\ \Rightarrow 0x + 3y - 30 &= 0 \\ \Rightarrow y &= 10 \end{aligned}$$

$$\begin{array}{c} | \\ \bullet \quad x=0 \\ | \end{array}$$

Starting : (5, 7)

$$dx = -5 (-) \quad |dx| > |dy|$$

Ending : (0, 10)

$$dy = 3 (+) \quad \boxed{\text{Zone: 3}}$$

Zone 3 $\rightarrow$ Zone 0

$$(5, 7) \rightarrow (-5, 7)$$

$$(0, 10) \rightarrow (0, 10)$$

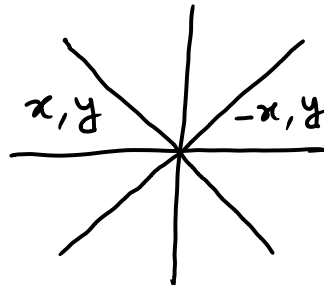
$$dx = 5 (+)$$

$$dy = 3 (+)$$

$$d_{init} = 2dy - dx = 1$$

$$\Delta d_{NE} = 2(dy - dx) = -4$$

$$\Delta d_E = 2dy = 6$$



$$d_{init} = 1 ; \Delta d_{NE} = -4 ; \Delta d_E = 6$$

x	y	d	NE/E	d_{update}	Zone 3 Pixels
-5	7	170	NE	$1 + \Delta d_{NE} = -3$	(5, 7)
-4	8	$-3 < 0$	E	$-3 + \Delta d_E = 3$	(4, 8)
-3	8	$3 > 0$	NE	$3 + \Delta d_{NE} = -1$	(3, 8)
-2	9	$-1 < 0$	E	$-1 + \Delta d_E = 5$	(2, 9)
-1	9	570	NE	$5 + \Delta d_{NE} = 1$	(1, 9)
0	10	170	NE	$1 + \Delta d_{NE} = -3$	(0, 10)

$$x = -X, y = Y$$

b) It is because, multiplying the decision parameter by 2 removes the fractional value, but it does not change the sign. The decision (NE/E) remains the same.